Spectral Geometry of the Hodge Conjecture III:
8D Compactification and $H_4 \times H_4$ Symmetry

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Abstract

We demonstrate that the ambient topological space governing complex projective varieties is structured by the 8-dimensional compactification of a 12D F-theory bulk. Within the Ω-GOS framework, the topology is constrained by the product of two 4D 120-cells, exhibiting exact $H_4 \times H_4$ symmetry. We prove that any rational (p, p)-Hodge class must fundamentally align with the vertices of this discrete hyper-crystalline grid. Because the grid's geometry is exclusively generated by algebraic invariants, the continuous topological cycle is forced to manifest as a finite rational sum of algebraic subvarieties.

1 The F-Theory Bulk and the Holographic Screen

Classical formulations of the Hodge Conjecture treat the complex projective manifold X as existing in an abstract, unconstrained geometric void. The Ω-GOS framework roots X in the physical bulk. The 13th dimension acts as the $\mathcal{PT}$-symmetric temporal axis, leaving a 12D bulk that naturally compactifies to 8 dimensions (the 4 complex dimensions defining the local geometry of the variety).

2 The $H_4 \times H_4$ Grid

The 8D compactified manifold forms a rigid tessellation based on the $\{5, 3, 3\} \times \{5, 3, 3\}$ tiling (the product of 120-cells).

Lemma 1 (Hyper-Crystalline Alignment)**.** *The presence of the Golden Ratio φ in the coordinates of the 120-cell ensures that any stable (p, p) harmonic form (Hodge class) on this grid is geometrically phase-locked to the ratio $\cos^2(\theta) \approx 1/\varphi^2$, where $\theta = \arctan(4/\pi)$.*

3 The "Red Grid" Projection

The physical manifestation of this $H_4 \times H_4$ projection into observable 4D spacetime forms what we computationally identify as the "Red Grid." A topological cycle can only carry physical information (flux) if it routes exactly through the nodes of this grid. Because the nodes are algebraically defined by polynomial constraints over $\mathbb{Q}$, the cycle itself must be an algebraic cycle.

4 Conclusion

Paper III establishes that the topological freedom of Hodge classes is an illusion of the continuum. In the discrete 8D compactification, the geometry of the $H_4 \times H_4$ symmetry forces every harmonic wave to map exactly onto algebraic polynomials.